A gap in the proof of Lemma 3.12 of Cavenagh–Greenhill–Wanless, *The cycle structure of two rows in a random Latin square*

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Abstract

Case 3 of the splitting procedure used to prove Lemma 3.12 of [2] (cross-switch with respect to $\{\omega(j), \omega(j')\}$, then backflip with respect to $\{j, j'\}$) does not behave as claimed: the cross-switch reverses an ω-arc containing j' (or j), after which $\{j, j'\}$ is at ω-distance 2, and when $\min(\alpha, \beta) \geq 3$ the backflip produces a square of the wrong type. The addition table of $\mathbb{Z}_7$ already shows this. When $\min(\alpha, \beta) = 2 < \max(\alpha, \beta)$ the type is right but the two multigraphs of the double count differ; for $n = 5$ we computed both exactly. The lower bound of Lemma 3.12 survives (and is elementary); the upper bound $3/2$ is not established by the published argument for any split other than $(2, 2)$, though all the evidence we have says that the statement is true, with room to spare. The body of the note records what can be recovered (the bound with constant 2 when $\min(\alpha, \beta) = 2$; a partial repair of case 3; a tail bound for the number of cycles from the surviving direction) and which later results use the affected estimates.

1 Setting and the procedure

We use the notation of [2], Section 3. L is a Latin square of order n containing a suitable partial square F; ω is the permutation of the first m columns with $1 \circ \omega(j) = 2 \circ j$, whose cycle type $\lambda \in P(m)$ defines $S(\lambda, F)$. For columns j, j' let $\delta_{j,j'}$ be the row permutation with $\delta(i) \circ j = i \circ j'$; $\{j, j'\}$ is an *A-pair* if rows $1, 2$ lie in different cycles of $\delta_{j,j'}$ and a *B-pair* otherwise. The *backflip* at an A-pair $\{j, j'\}$ lying on one ω-cycle of length $\alpha + \beta$ with $\omega^\alpha(j) = j'$ trades the $\delta_{j,j'}$-cycle through row 2; it splits that ω-cycle into the two arcs between j and j', of lengths α and β (Lemma 3.5). The same trade at an A-pair whose columns lie on different ω-cycles, of lengths α and β, is the *flip* (Definition 3.2); it joins the two cycles into one, with $\omega'^\beta(j) = j'$ (Lemma 3.3). The *switch* at such a pair is the trade of rows $1, 2$ along one of the two ω-cycles (Definition 3.6 selects the cycle containing the smallest symbol; below we sometimes fix the cycle by another rule, which is harmless for the counts), and it turns a B-pair into an A-pair (Lemma 3.7). The *cross-switch* at a B-pair $\{j, j'\}$ with $\omega^\alpha(j) = j'$ (Definition 3.9) is a Latin bitrade (Q, Q') that preserves $S(\lambda, F)$ (Lemma 3.10).

Let μ be λ with one part $\alpha+\beta$ split into $\alpha, \beta \geq 2$. The splitting procedure (p. 294) considers, for $L \in S(\lambda, F)$, every pair $\{j, j'\}$ with $\omega^\alpha(j) = j'$ on an $(\alpha+\beta)$-cycle and produces $L' \in S(\mu, F)$ as follows:

1. if $\{j, j'\}$ is an A-pair, backflip at $\{j, j'\}$;

2. else if $\{\omega j, \omega j'\}$ is an A-pair, backflip at $\{\omega j, \omega j'\}$;

3. otherwise cross-switch with respect to $\{\omega j, \omega j'\}$; "now $\{j, j'\}$ has become an A-pair, and we backflip with respect to $\{j, j'\}$".

Item 4 asserts that in each case $L' \in S(\mu, F)$. Together with a fifth edge (a switch) this gives every $L \in S(\lambda, F)$ degree exactly $2(\mu_{\alpha+\beta}+1)(\alpha+\beta)$ in a bipartite multigraph between $S(\lambda, F)$ and $S(\mu, F)$, while the joining procedure gives every $L \in S(\mu, F)$ degree between $\mu_\alpha \mu_\beta \alpha\beta$ and three times that; the two multigraphs are asserted to coincide, and the double count is Lemma 3.12:

$$\frac{1}{2} \;\le\; \frac{|S(\lambda,F)|\,(\mu_{\alpha+\beta}+1)(\alpha+\beta)}{|S(\mu,F)|\,\mu_\alpha\mu_\beta\,\alpha\beta} \;\le\; \frac{3}{2} \qquad (\alpha \neq \beta). \tag{1}$$

(For $\alpha = \beta$ the statement has $2(\mu_{2\alpha}+1)/(\alpha\mu_\alpha(\mu_\alpha-1))$ in place of the fraction; everything below applies verbatim, with the splits (α,α), $\alpha \ge 3$, in the class $\min(\alpha,\beta) \ge 3$ and $(2,2)$ the only unaffected split.)

2 The cross-switch reverses an arc

Write $e_1 = 1 \circ j$, $e_2 = 1 \circ j'$, $e_3 = 2 \circ j$, $e_4 = 2 \circ j'$ and consider the first form of Definition 3.9 ($\min(e_1, e_2, e_3, e_4) \in \{e_1, e_4\}$); the second form ($\min \in \{e_2, e_3\}$) is the first form applied to the ordered pair (j', j), i.e. with the roles of j and j' (and of α and β) exchanged, so that the cross-switch at an unordered B-pair is well defined. On rows $1, 2$ the bitrade is

$$Q' \supseteq \{(2, j, e_2), (1, j', e_3)\} \cup \{(1, \omega^k j,\ 2 \circ \omega^k j),\ (2, \omega^k j,\ 1 \circ \omega^k j)\ :\ 1 \le k < \alpha\},$$

i.e. in every interior column of the ω-arc from j to j' the entries of rows $1, 2$ are swapped, row 2 receives $e_2 = 1 \circ j'$ at column j, and row 1 receives $e_3 = 2 \circ j$ at column j'. (This is the definition; the proof of Lemma 3.10 confirms it: row 1 keeps the symbol set $\{1 \circ \omega^k j : 1 \le k \le \alpha\}$ on the cells $\{\omega^k j : 1 \le k < \alpha\} \cup \{j'\}$, and "precisely one column from the pair $\{\omega^k j, \omega^{k'} j'\}$ has symbols swapped in rows 1 and 2".) The remaining cells of Q lie in columns j, j' and in rows other than $1, 2$; they do not affect ω.

Lemma 1. *Let L' be the cross-switch of L at the B-pair $\{j, j'\}$, $\omega^\alpha(j) = j'$, and let $\omega' = \omega_{L'}$. Then $\omega'(j) = \omega^{\alpha-1}(j)$, $\omega'(\omega^k j) = \omega^{k-1}(j)$ for $2 \le k < \alpha$, $\omega'(\omega j) = j'$, and $\omega' = \omega$ on every other column. In words: the interior of the ω-arc from j to j' is traversed in reverse, and the cycle structure of ω is otherwise unchanged.*

Proof. Write $1', 2'$ for the new rows. Then $1' \circ j = e_1$, $1' \circ \omega^k j = 1 \circ \omega^{k+1} j$ for $1 \le k < \alpha$ (since $2 \circ \omega^k j = 1 \circ \omega^{k+1} j$), $1' \circ j' = 1 \circ \omega j$; $2' \circ j = 1 \circ j' = 1 \circ \omega^\alpha j$, $2' \circ \omega^k j = 1 \circ \omega^k j$ for $1 \le k < \alpha$, $2' \circ j' = e_4$. Now $\omega'(c)$ is the column whose row-$1'$ entry equals $2' \circ c$: for $c = j$ this is $\omega^{\alpha-1} j$; for $c = \omega^k j$ with $2 \le k < \alpha$ it is $\omega^{k-1} j$; for $c = \omega j$ it is j' (the symbol $1 \circ \omega j$ now sits at $(1', j')$); for $c = j'$ it is $\omega j'$, unchanged; for $c = \omega^{-1} j$, $2' \circ c = 2 \circ c = e_1 = 1' \circ j$, so $\omega'(c) = j$; all other columns are untouched in both rows. □

The example on p. 293 of [2] shows exactly this: there $\omega = (1\,2\,3\,4\,5\,6\,7)$, $j = 2$, $j' = 5$, $\alpha = 3$, and the displayed cross-switch has rows $1\,2\,4\,5\,3\,6\,7$ and $2\,5\,3\,4\,6\,7\,1$, i.e. $\omega' = (1\,2\,4\,3\,5\,6\,7)$.

3 Case 3 lands in the wrong type

Proposition 2. *Let $\{j, j'\}$ and $\{J, J'\} = \{\omega j, \omega j'\}$ both be B-pairs with $\omega^\alpha j = j'$ on an $(\alpha+\beta)$-cycle, and let L' be the cross-switch of L at $\{J, J'\}$ (either form). Then $\{j, j'\}$ is an A-pair of L', but in the first form it is at ω'-distance 2 (from j to j'), and in the second form at ω'-distance 2 from j' to j. Consequently the backflip at $\{j, j'\}$ splits the $(\alpha + \beta)$-cycle into parts 2 and $\alpha + \beta - 2$, which is the type μ only if $\min(\alpha, \beta) = 2$.*

Proof. First form. The arc from $J = \omega j$ to $J' = \omega^\alpha J$ has interior $\omega J, \ldots, \omega^{\alpha-1} J$, and $\omega^{\alpha-1} J = \omega^\alpha j = j'$. By Lemma 1, $\omega'(j) = \omega'(\omega^{-1} J) = J$ and $\omega'(J) = \omega^{\alpha-1} J = j'$. The status of $\{j, j'\}$: column j' is an interior column of the arc, so rows $1, 2$ are swapped there, while column j is untouched ($j \notin \{J, J'\}$ and j is not in the arc); hence $\delta_{j,j'}$ changes by a transposition of rows $1, 2$ and $\{j, j'\}$ toggles from B to A, as claimed in [2]. The second form reverses the arc from J' to J, whose interior contains $j = \omega^{-1} J$ as its last column, and the same computation gives $\omega'(J') = j$, i.e. $\omega'^2(j') = j$. □

So the asserted membership $L' \in S(\mu, F)$ in item 4 fails in case 3 whenever $\min(\alpha, \beta) \geq 3$.

An example by hand. Take the addition table of $\mathbb{Z}_7$, $L(r, c) = r + c \bmod 7$, with the rows indexed by 0 and 1 in the roles of rows 1 and 2 (and $F = \emptyset$). Then $\omega(c) = c + 1$, a single 7-cycle. Take $j = 0$, $j' = 3$, $(\alpha, \beta) = (3, 4)$. For any two columns c, c' the permutation $\delta_{c,c'}$ is the translation $r \mapsto r + c' - c$, a single cycle when $c \neq c'$, so every pair is a B-pair: $\{0, 3\}$ and $\{\omega 0, \omega 3\} = \{1, 4\}$ in particular, and the procedure is in case 3. The corner symbols at $\{1, 4\}$ are $e_1 = 1$, $e_2 = 4$, $e_3 = 2$, $e_4 = 5$, so the cross-switch takes its first form and reverses the interior $2, 3$ of the arc from 1 to 4: the new rows are $0\,1\,3\,4\,2\,5\,6$ and $1\,4\,2\,3\,5\,6\,0$, with $\omega' = (0\,1\,3\,2\,4\,5\,6)$ and $\omega'^2(0) = 3$. The pair $\{0, 3\}$ is now an A-pair, as item 3 says, but at distance 2; the backflip at $\{0, 3\}$ gives type $(2, 5)$, not $(3, 4)$.

The case $\min(\alpha, \beta) = 2$. Say $\alpha = 2 \leq \beta$. In the form of the cross-switch that reverses the arc of length 2 (a single interior column), nothing is reversed and case 3 is as described. In the other form the arc of length β is reversed; j or j' is its last interior column, the backflipped square has parts 2 and β, so it does lie in $S(\mu, F)$, and item 4 holds. But the proof of Lemma 3.12 rests on the identity $G_S = G_J$, and this fails: the case-3 edge L–L' is supposed to be the joining rule-4 edge from L' (flip at $\{j, j'\}$ to L'', then cross-switch at $\{\omega_{L''} j, \omega_{L''} j'\}$), which returns L by Corollary 3.11 only if $\{\omega_{L''} j, \omega_{L''} j'\} = \{J, J'\}$; and when the long arc has been reversed this is not so (e.g. in the second form $\omega_{L''}(j) = \omega^{\beta-2} J' \neq J$). Which form occurs is decided by the position of the smallest corner symbol, not by the author, so about half of the case-3 edges and their switch edges (item 5) are missing from G_J. We computed both multigraphs exactly for $n = 5$, $\lambda = (5)$, $\mu = (2, 3)$, $F = \emptyset$, over all 161280 Latin squares of order 5 (the procedures select trades by the smallest symbol, so one cannot normalise the first row): G_S and G_J each have 1036800 edges, but 172800 edges of G_S are not edges of G_J (exactly half of the case-3 edges and the switch edges of the same instances; all edges of cases 1 and 2 are in G_J), and the degrees of the μ-squares in G_S are 12, 18 and 24 (for 11520, 34560 and 11520 squares) against 18 for every μ-square in G_J. So the per-vertex assertion "each vertex of $S(\mu, F)$ has degree $d \leq 3\mu_\alpha \mu_\beta \alpha\beta$" is false for the splitting multigraph, though it holds on average over $S(\mu, F)$; the inequality $Z \leq 3\mu_\alpha \mu_\beta \alpha\beta |S(\mu, F)|$ itself therefore still holds at $n = 5$, with equality (the ratio is exactly 3/2 there), but not for the reason given. For $(4) \to (2, 2)$ at $n = 4$ the two multigraphs

coincide, as they must: only arcs of length 2 occur. In summary, the published argument for the upper bound is incomplete for every split with $\max(\alpha,\beta)\geq 3$.

Numerical confirmation. We implemented Definition 3.9 literally (rows $1,2$ as above; in columns j,j' the rows $\delta^k(1)$, $1\leq k<a$, with $\delta^a(1)=2$, swap their two entries, exactly as in the definition — this is what balances the corner cells) and checked on Latin squares produced by the Jacobson–Matthews chain [6] (not certified uniform; the correctness checks below do not depend on their distribution, the estimates of §8 do) that the cross-switch is a Latin square of the same type in every instance. Applying case 3 as written: in 1006 ordered instances (j,j') with $\min(\alpha,\beta)\geq 3$ at $n=9$ (300 sampled squares) and 3098 at $n=12$ (300 squares), the result had the wrong type in every single instance (e.g. $(2,7)$ with $(\alpha,\beta)=(3,4)$ gives $(2,2,5)$ instead of $(2,3,4)$; (9) with $(3,6)$ gives $(2,7)$).

4 What survives, and what depends on the gap

Only case 3 is affected. The joining rules 1–2 are valid and each rule-2 edge is a case-1 edge or the switch edge of a case-1 instance, so each $L\in S(\mu,F)$ has degree at least $\mu_\alpha\mu_\beta\alpha\beta$ in the multigraph of valid splitting edges, in which each $L\in S(\lambda,F)$ has degree at most $2(\mu_{\alpha+\beta}+1)(\alpha+\beta)$. Hence the *lower* bound $\frac{1}{2}$ in (1) stands. (In fact it is elementary: with X the set of pairs $(L,\{j,j'\})$, $L\in S(\lambda,F)$, $\omega^\alpha j=j'$, and J the analogous set of joining pairs, the flip and the switch give $|J|=2|X_A|$ exactly, so (1) reads $\frac{1}{3}\leq|X_A|/|X|\leq 1$, and the lower bound $\frac{1}{2}$ is $|X|\geq|X_A|$.) The *upper* bound $\frac{3}{2}$, equivalently

$$\Pr\big[\{j,j'\}\text{ is an }A\text{-pair}\big]\;\geq\;\tfrac{1}{3}\quad\text{for }(L,\{j,j'\})\text{ uniform on }X,$$

is not established by the published argument for $\max(\alpha,\beta)\geq 3$; Proposition 3 below establishes it with $\frac{1}{4}$ in place of $\frac{1}{3}$ for $\min(\alpha,\beta)=2$; for $\min(\alpha,\beta)\geq 3$ we know of no proof.

Within [2] the upper bound of Lemma 3.12 is used for: the upper bound of Theorem 3.13; the lower bound $(2/3)^{\kappa-1}$ of Lemma 4.1 and, in Lemma 4.2, the upper bound whenever the path from λ to μ contains a splitting step and the lower bound whenever it contains a joining step; the bounds on $\mathbb{P}_F(E)$ of Theorem 4.3 (whose proof invokes the equality of the two multigraphs "as in the proof of Lemma 3.12"); the upper bound $\mathbb{P}_F((m))\leq 2m^{-2/3}$ of Lemma 4.4; both halves of Corollary 4.5 (its lower half uses the $(2/3)^{\kappa-1}$ bound, its upper half the factor $m^{1/3}$ from Lemma 4.4); and the results that rest on these: Corollary 4.6, Lemma 4.7, Theorem 4.9, Corollary 4.10 and Theorem 4.11. (Corollary 4.10, on the number of intercalates, is in any case superseded by Kwan and Sudakov [8].) The upper bound $2^{\kappa-1}$ of Lemma 4.1, the lower bound $\mathbb{P}_F((m))\geq 2m^{-2}$ of Lemma 4.4, Corollary 4.8, and the upper bounds on short-cycle counts derived from the joining direction are unaffected. For instance, from the joining direction alone one still gets $\mathbb{E}[\ell N_\ell]\leq 2m/(m-\ell)$ for the number N_ℓ of ℓ-cycles, $\ell<m$ (sum the inequality $\mathbb{E}[\ell N_\ell\,\beta N_\beta]\leq 2\mathbb{E}[(\ell+\beta)N_{\ell+\beta}]$, with $N_\ell N_\beta$ read as $N_\ell(N_\ell-1)$ when $\beta=\ell$, over β), and by joining k cycles successively one obtains a factorial-moment bound of the form $\mathbb{E}[(\kappa)_k]\leq 2^{k-1}(4+k^2/\log m)(\log m)^k$ for the number of cycles κ, valid for every k (the derivation is in the Appendix). With $k=\log m$ this gives $\Pr[\kappa\geq 16\log m]=O(m^{-2}\log m)$, and with $k=\lceil\sqrt{m}\rceil$ it gives, by Markov's inequality on $(\kappa)_k\geq(8\sqrt{m})^k$,

$$\Pr[\kappa\geq 9\sqrt{m}]\leq m\Big(\frac{\log m}{4\sqrt{m}}\Big)^{\sqrt{m}}=\exp\big(-(\tfrac{1}{2}-o(1))\sqrt{m}\,\log m\big),$$

which is the second assertion of Theorem 4.9 with a better rate. The first assertion, $\Pr[\kappa \ge k] = o((\frac{3}{4})^k \exp(\pi\sqrt{2m/3}))$, also follows: it is vacuous unless $k \ge 8.9\sqrt{m}$, and for such k Markov's inequality on $(\kappa)_{k'}$ with $k' = \lceil k/2 \rceil$ gives $\Pr[\kappa \ge k] \le \frac{1}{2}(4 + k'^2/\log m)(4\log m/k)^{k'}$, which is $o((\frac{3}{4})^k)$. So the statements of Theorem 4.9 and of the $\log m$-scale tail survive (by a different proof), while the lower bound on the number of cycles of Theorem 4.11 does not. Likewise the upper half of Corollary 4.5 survives in a weaker form. Write $\mathcal{C}(\lambda) = |S(\lambda, F)|/\gamma(\lambda)$, where $\gamma(\lambda)$ is the number of permutations of type λ, so that $\mathcal{C}(\lambda)/m!$ is the number of completions per two-row rectangle of type λ compatible with F (there are $m!\gamma(\lambda)$ such rectangles; for $F \ne \emptyset$ the individual counts may differ and this is their average; the factor $m!$ cancels from every ratio below) and (1) reads $\frac{1}{2} \le \mathcal{C}(\lambda)/\mathcal{C}(\mu) \le \frac{3}{2}$. Joining the κ parts of a type one pair at a time and using at each step the surviving inequality $\mathcal{C}(\text{finer}) \le 2\,\mathcal{C}(\text{coarser})$ (valid also when the two joined parts are equal) gives $\mathcal{C}(\lambda) \le 2^{\kappa-1}\mathcal{C}((m))$ for a type λ with κ parts (this is the upper half of Lemma 4.1, listed above as unaffected), hence

$$\mathbb{P}_F(\lambda) \le \frac{\gamma(\lambda)\,2^{\kappa-1}\mathcal{C}((m))}{\gamma((m))\,\mathcal{C}((m))} \le \frac{m}{2}\prod_{i=1}^{\kappa}\frac{2}{c_i}$$

for a type with parts $c_1, \dots, c_\kappa$, using $\gamma(\lambda) \le m!/\prod c_i$, $\gamma((m)) = (m-1)!$ and $\mathbb{P}_F(\lambda) \propto \gamma(\lambda)\mathcal{C}(\lambda)$ (in place of the factor $m^{1/3}$, which came from Lemma 4.4).

The surviving direction bounds fine types from above only: it gives no upper bound on the probability of a coarse type. In particular we know of no proof, from the surviving material or from our own identities, that the first two rows of a random Latin square form a single n-cycle with probability $o(1)$ (the statement $\mathbb{P}_F((m)) \le 2m^{-2/3}$ of Lemma 4.4). What is available, for $F = \emptyset$, is a constant: by row-pair symmetry (which conditioning on $F \ne \emptyset$ destroys, so the following is stated for $F = \emptyset$ only, $m = n$) and the intercalate count of [8], $\mathbb{E}N_2 \ge \frac{1}{2} - o(1)$; the $\ell = \beta = 2$ case of the surviving inequality gives $\mathbb{E}[N_2(N_2-1)] \le 2\mathbb{E}N_4 \le m/(m-4)$; so by the second moment method $\Pr[N_2 \ge 1] \ge (\mathbb{E}N_2)^2/\mathbb{E}[N_2^2] \ge \frac{1}{6} - o(1)$ and $\mathbb{P}_\emptyset((n)) \le \frac{5}{6} + o(1)$. Proposition 3 does not improve this: it gives $\mathbb{P}_F((m)) \le 4\mathbb{P}_F((2, m-2))$, the surviving inequality gives $\mathbb{P}_F((2, m-2)) \le \frac{m}{m-2}\mathbb{P}_F((m))$, and iterating over k two-cycles loses a factor of order $4^k k!$.

One statement in the coarse direction is available from a different source, the switchings of Allsop and Morris [1]. Their Claims 2–3 (the cross-switch count of good pairs on a column cycle, and the double count of row switches that cut pieces off the cycles through the special rows, closed by Jensen's inequality) can be run with a single special row x and no fixed cells, and give, for a uniform Latin square and a fixed column pair, $\mathbb{E}[n - \ell_x] \ge (n-1)/30$ for the length ℓ_x of the cycle through x (for $n \ge 502$, with the constants of [5, §7]). By row exchangeability this is $\Pr[x, y \text{ on the same cycle}] \le 1 - \frac{1}{30}$ for two fixed rows, and summing over x, with M the length of the longest cycle of the pair and $\sum_C |C|(n - |C|) \le 2n(n-M)$,

$$\mathbb{E}[n - M] \;\ge\; \frac{n-1}{60}, \qquad \Pr\Big[n - M \ge \frac{n}{120}\Big] \;\ge\; \frac{1}{120} - \frac{1}{60n};$$

by transposition the same holds for the longest cycle of a fixed row pair, i.e. for $\mathbb{P}_F$: with probability at least $1/120 - o(1)$ the columns off the longest cycle of $\rho_{1,2}$ are at least $n/120$ in number. This bounds the probability of the union of the coarsest types (longest part $> (1 - 1/120)m$; here $n = m$) away from 1, which the surviving material does not do; it is weaker than $5/6$ for the type (m) itself and says nothing about $\mathbb{P}_F((m)) = o(1)$. (The argument, with the two rows of a fixed rectangle as further special rows, is written out in [5, §8]; apart from the distance-2 case of Proposition 3 it is the only lower bound on a "different cycles" event we know, and it uses the exchangeability of the second row in an essential way.)

5 Other affected papers

We went through the titles of the papers citing [2] that are indexed by MathSciNet or listed by Google Scholar and read those whose titles suggested a use of the affected results: besides [4, 3, 7], these were Kwan–Sah–Sawhney–Simkin, Allsop–Wanless, Divoux–Kelly–Kennedy–Sidhu and Allsop–Morris [1], which cite [2] for the intercalate estimates, superseded by [8], or, in the last case, for the reversibility and effect lemmas for cycle and cross switches in §3 of [2], which are unaffected. We have not read the remaining citing papers, whose titles concern autotopisms, transversals, bitrades and loops. Two of the papers read use the affected results in an essential way; one of them is itself used by a third.

Gill, Mammoliti and Wanless [4] use Corollary 4.5 (upper half) and Theorem 4.9 to prove their Theorem 9 (with probability $1 - o(\exp(-\frac{1}{4}n^{1/2}))$ the first two rows of a random Latin square contain a row cycle of length at least $n^{1/2}\log^{1-\varepsilon} n$), on which the average-case bound of their Theorem 1 rests through Theorem 12. Both inputs are affected by the gap, but both are replaced by the surviving bounds just stated: with $\mathbb{P}_F \le \frac{m}{2}\prod(2/c_i)$ in place of $m^{1/3}\prod(2/c_i)$ the estimate in their proof becomes $\exp(-\delta\sqrt{n}(1-o(1))\log^{\varepsilon} n)$ times a polynomial factor, and the number of cycles is at most $9\sqrt{n}$ with probability $1 - \exp(-\Omega(\sqrt{n}\log n))$; so their Theorem 9 holds as stated, granting the derivation in the Appendix.

Cavenagh and Wanless [3] use three results of [2] in the proof of their Theorem 1.1 (asymptotically the same number of Latin squares of each parity): the upper half of Corollary 4.5 in their Lemma 2.1 (no row cycle of length $\ge n - \log n$ in the last two rows, with probability $1 - o(1)$), Theorem 4.9 in their Lemma 2.2, and the upper bound of Theorem 3.13 in their Lemma 2.3, for the splits of an even part z into two odd parts $a, z-a$ (so $\min(a, z-a) \ge 3$). The second use is repaired as above (the Appendix is stated for $F = \emptyset$; its proof uses only flips and switches at marks in the free columns and trades along cycles outside F, so it applies in $\Omega(F)$ with m in place of n, but we have not written this out; their Lemma 2.2 states the probability as $1 - o(c^m)$, which neither Theorem 4.9 nor the repair gives, but only $1 - o(1)$ is used). The first is not repaired by the weaker surviving constant: with $n/2$ in place of $n^{1/3}$ the estimate in Lemma 2.1 becomes $O(2^{(\log n)/2}\log n)$, and already its $\lambda = (n)$ term is 1; what Lemma 2.1 needs is that the cycle through column 1 has length at most $n - \omega(1)$ with probability $1 - o(1)$, which includes $\mathbb{P}_F((n)) = o(1)$, which, as noted at the end of §4, we cannot prove either. The third is the unproved direction at exactly the splits where case 3 fails, outside the range of Proposition 3; what their Lemma 2.3 actually needs is weaker than $\frac{3}{2}$ — any constant, or even the averaged statement $\sum_{\mu\in M}\gamma(\mu)|S(\mu,F)|/(\gamma(\lambda)|S(\lambda,F)|) \to \infty$ over the odd splits M of a large even part — but we do not have that either. So, as far as we can see, the published proof of Theorem 1.1 of [3] uses the affected estimates at two points for which this note supplies no replacement. The statement is expected to be true independently of the gap: Kwan, Petrova and Sawhney [7] have since proved Cameron's conjecture on parities (their Theorem 1.2, and the trivariate Theorem 1.3), of which Theorem 1.1 of [3] is a special case (Theorem 1.3(4), as they note); their written proof of part (3), hence of (4), uses [3] as a black box, so formally the dependence on the gap runs through it, but their Remark 6.6 sketches an independent re-proof of the Cavenagh–Wanless theorem by their own method (switching a 2×3 and a 3×2 subrectangle), which we have not checked. Independently of this, their large deviation principle, Theorem 1.3(5), is deduced from their Theorem 6.4 and Lemma 6.5(ii) without any use of [3] or of [2], and it implies the law of large numbers for the number of odd rows (their Theorem 1.3(1)), hence, as they remark, by row exchangeability $\Pr[\text{rows } 1,2 \text{ have the same parity}] = \frac{1}{2} + o(1)$; this supersedes Theorem 4.3 of [2] (affected by the gap, as listed above). (Their written deduction of the law of large numbers

goes through part (4), hence through [3]; the route through part (5) avoids it.)

6 A partial repair

A cross-switch at a B-pair $\{J, J'\}$ at any ω-distance a with $2 \le a \le t-2$, t the length of the containing cycle (both arcs must have an interior), is a type-preserving trade (the proof of Lemma 3.10 does not use $a = \alpha$), and by Lemma 1 it reverses the arc from J to J'. If $\{J, J'\}$ is disjoint from $\{j, j'\}$, it toggles the status of $\{j, j'\}$ iff exactly one of j, j' is an interior column of that arc, say at position i ($1 \le i < a$), and then the ω-distance from j to j' changes by $\pm(a - 2i)$ (sign according to which of j, j' is interior). Thus, among cross-switches at pairs disjoint from $\{j, j'\}$, the status-toggling moves that preserve the distance are precisely those at B-pairs *symmetric about j or about j'* ($a = 2i$).

Suppose α is even and $\alpha < 2\beta$, and let $R = \{\omega^{\alpha/2} j,\ \omega^{3\alpha/2} j\}$: the pair at distance α symmetric about j', whose arc avoids j. If R is a B-pair, cross-switch at R: $\{j, j'\}$ becomes an A-pair whose two ω'-arcs still have lengths α and β (in the form that reverses the arc of R through j' the distance from j to j' stays α; in the other form it becomes β), and the backflip at $\{j, j'\}$ lies in $S(\mu, F)$. If R is an A-pair, backflip at R itself, which is a pair at distance α. Either way case 3 acquires a valid edge (verified on 3928 instances in Jacobson–Matthews samples at $n = 10, 13$: type μ in all of them). The same works with the roles of α, β exchanged when β is even and $\beta < 2\alpha$. The multiplicities on the $S(\mu, F)$ side change (an A-pair's backflip may now serve two splitting pairs), so the constant $3/2$ would have to be recomputed.

When α and β are both odd, every distance-preserving toggle has even arc length, so no symmetric pair is at distance α or β; if all symmetric pairs about j and j' happen to be A-pairs, no single cross-switch repairs $\{j, j'\}$, and two-step repairs pass through a third type, which breaks the two-sided count. We do not see a canonical repair in this case (nor when $\alpha \ge 2\beta$ with β odd). Repairs that use other rows do exist — for a B-pair $\{j, j'\}$ with column cycle $Z \ni 1, 2$, trade rows z, w lying on different arcs of Z between 1 and 2 along the row cycle of $\{z, w\}$ through j (if j, j' lie in different row cycles of $\{z, w\}$; this separates 1 from 2 and keeps the type), or cross-switch rows z, w at $\{j, j'\}$ (if they lie in one; this changes the type to μ) — but they are indexed by a free pair of rows and so have unbounded multiplicity on the $S(\mu, F)$ side, which defeats the double count.

7 The case $\min(\alpha, \beta) = 2$: the bound holds with constant 2

For $\min(\alpha, \beta) = 2$ the published count can be corrected at the price of a weaker constant. Say $\alpha = 2 \le \beta$ and write $t = \alpha + \beta = \beta + 2$ for the length of the part being split (m keeps its meaning, the number of free columns).

Proposition 3. *If $\alpha = 2$ (or $\beta = 2$), then*

$$\frac{|S(\lambda, F)|\,(\mu_t + 1)\,t}{|S(\mu, F)|\,\mu_2 \mu_\beta\, 2\beta} \ \le\ 2 \qquad (\beta \ne 2),$$

and for $\alpha = \beta = 2$ the quantity of Lemma 3.12 is ≤ 2 as well.

Proof. Every splitting instance $(L, \{j, j'\})$ yields two squares in $S(\mu, F)$: in cases 1 and 2 this is item 4, and in case 3 the cross-switch either reverses nothing (the arc of length 2) or reverses the

arc of length β, and in both forms the backflip at $\{j, j'\}$ produces parts 2 and β (Proposition 2). So $Z = 2(\mu_t + 1)t\,|S(\lambda, F)|$ counts valid edges (for $\alpha = \beta = 2$, where pairs are unordered, $Z = 2(\mu_t + 1)(t/2)|S(\lambda, F)|$). It remains to show that every $M \in S(\mu, F)$ receives at most 4 edges per joining pair, i.e. per pair $\{q, q'\}$ with q in a 2-cycle and q' in a β-cycle of ω_M. Attribute each edge (L, M) to a joining pair of M as follows: the direct edge of case $c \in \{1, 2, 3\}$ to the pair at which the backflip was made ($\{j, j'\}$ in cases 1, 3; $\{\omega j, \omega j'\}$ in case 2), which is an A-pair of M (the backflipped pair is an A-pair afterwards); the switch edge of cases 1 and 3 to the switched pair $\{j, j'\}$, which is a B-pair of M (Lemma 3.7: it was an A-pair before the switch); and the switch edge of case 2 to $\{\omega j, \omega j'\}$, which is a B-pair of M (it was the backflipped A-pair of L', and the switch at $\{j, j'\}$ trades the row cycle through j or through j', which contains exactly one of $\omega j, \omega j'$, so the status toggles by the argument in the proof of Lemma 3.7). Now fix M and a joining pair Q of M and count the edges attributed to Q.

If Q is an A-pair of M, only direct edges arrive, and $L_1 := \mathrm{flip}_Q(M)$ is determined. Case 1 contributes (L_1, Q); case 2 contributes $(L_1, \{\omega_{L_1}^{-1} q, \omega_{L_1}^{-1} q'\})$; in case 3, L is a cross-switch of L_1 at a B-pair $\{J, J'\}$ with $Q = \{j, j'\}$ at ω_L-distance 2, and $\{J, J'\}$ is determined by L_1 and Q: let (a, b) be the ordering of Q with $\omega_{L_1}^2(a) = b$ (unique since $\beta \geq 3$; for $\beta = 2$ both orderings give the same candidate below and there is no long arc, so case 3 then contributes at most 1); if nothing was reversed, $\{J, J'\} = \{\omega_{L_1} a, \omega_{L_1} b\}$; if the long arc was reversed, the computation of Proposition 2 (with j' the column of Q at distance 2 after j in L) gives $\omega_{L_1}(j') = J'$ and $\omega_{L_1}^{-1}(j') = J$, i.e. $(a, b) = (j', j)$ and $\{J, J'\} = \{\omega_{L_1}^{-1} a, \omega_{L_1} a\}$, the pair symmetric about a. So case 3 contributes at most 2 edges. Total ≤ 4.

If Q is a B-pair of M, only switch edges arrive. For cases 1 and 3 the switch was made at Q itself, $M' = \mathrm{sw}_Q(M)$ is determined (the switch is an involution: it preserves the symbol sets of the two row cycles, so the smallest-symbol rule re-selects the same cycle), and the direct edge into M' at the A-pair Q is the case-1 edge or one of the two case-3 edges listed above: ≤ 3 edges. For case 2, $Q = \{\omega j, \omega j'\}$ and the switch was made at $\{j, j'\}$; the switch inverted the row cycle of M' through j or through j' (whichever carries the smaller symbol, which is read off from M since the symbol sets are preserved), so $j = \omega_M(\omega j)$ if the cycle through j was inverted and $j = \omega_M^{-1}(\omega j)$ otherwise, and similarly j'; then $M' = \mathrm{sw}_{\{j,j'\}}(M)$ and $L = \mathrm{flip}_Q(M')$ are determined: 1 edge. Total ≤ 4.

Hence $Z \leq 4\,\mu_2 \mu_\beta\, 2\beta\,|S(\mu, F)|$ (resp. $4\binom{\mu_2}{2} 4 |S(\mu, F)|$ for $\alpha = \beta = 2$), and the result follows as in the proof of Lemma 3.12. □

We verified the recipe by computer: on Jacobson–Matthews samples at $n = 7, 10, 12$ every direct and every switch edge of every splitting instance with $\alpha = 2$ is recovered from its target and attributed pair by the inverse recipes above, with at most 4 candidates per pair (`s13_cgw_min2.py`); and in the exact $n = 5$ computation of Section 3 the multiplicities per (target, attributed pair) are exactly 2, 3 or 4, direct edges always arrive at A-pairs and switch edges at B-pairs. The constant 2 is what this count gives at $n = 5$ ($24 = 4 \cdot 6$ is attained), where the truth is 3/2.

In the language of Section 4, Proposition 3 says that for a marked pair at distance 2 (or $t - 2$) on a t-cycle, $\Pr_X[A] \geq \frac{1}{4}$: a lower bound on a different-cycles event, which the surviving direction alone could not give. The reason it works for $\alpha = 2$ and not for $\min(\alpha, \beta) \geq 3$ is not multiplicity but type: for $\min(\alpha, \beta) \geq 3$ the wrong-form case-3 edges land in $S(\mu', F)$ with μ' the $(2, t - 2)$ split, so the same count gives only $2(|X| - |X_3|) \leq 3|J|$, where X_3 is the set of instances in case 3 ($\{j, j'\}$ and $\{\omega j, \omega j'\}$ both B-pairs), i.e. $\Pr_X[A] \geq \frac{1}{3}(1 - \Pr_X[\text{case 3}])$, which is empty without a lower bound on $\Pr_X[A \text{ at } \{j, j'\} \text{ or at } \{\omega j, \omega j'\}]$.

8 What the data say

The statement of Lemma 3.12 is almost certainly true, and far from sharp for $n \geq 6$. By the identity $|J| = 2|X_A|$ the quantity in (1) is $\mathcal{C}_n(\lambda)/\mathcal{C}_n(\mu)$, where $\mathcal{C}_n(\lambda)$ is the number of completions of two rows of type λ, and $\Pr_X[A] = \mathcal{C}_n(\mu)/(2\mathcal{C}_n(\lambda))$. From the exact values of $\mathcal{C}_n(\lambda)$ tabulated in [2] (Tables 1 and 2, $n \leq 11$):

n	#splits ($\lambda \to \mu$)	$\max \mathcal{C}_n(\lambda)/\mathcal{C}_n(\mu)$	$\min \Pr_X[A]$
5	1	1.5000 (sharp)	0.3333
6	3	0.9545	0.5238
7	4	1.0035	0.4982
8	8	1.0017	0.4991
9	11	1.0002	0.4999
10	19	1.0000	0.5000
11	26	1.0000	0.5000

(all splits with $\alpha, \beta \geq 2$; for $n \geq 7$ every ratio lies in $[0.985, 1.0035]$, with the minima 0.9861 at $(3,4) \to (2,2,3)$, $n = 7$, and 0.9853 at $(2,2,4) \to (2,2,2,2)$, $n = 8$, and the excess above 1 at most 0.35%, in the covered and the uncovered classes alike — e.g. $(2,6) \to (2,3,3)$ at $n = 8$, in the $(3,3)$ class, has ratio 1.0017). On squares produced by the Jacobson–Matthews chain, with a uniformly random marked pair at distance $\alpha \geq 2$ on a t-cycle, $t - \alpha \geq 2$, we find $\Pr_X[B] - \Pr_X[A] = -0.022 \pm 0.011$, $+0.008 \pm 0.012$, -0.030 ± 0.018 at $n = 30, 50, 100$ (9000, 7200, 3600 instances), i.e. $\Pr_X[A] = 0.50 \pm 0.01$ throughout. So the true constant in the upper bound of (1) appears to be $1 + o(1)$ (the maximal excess over 1 in the table is $3.5 \cdot 10^{-3}$, $1.7 \cdot 10^{-3}$, $1.8 \cdot 10^{-4}$, $< 10^{-5}$, $< 10^{-5}$ for $n = 7, \ldots, 11$) rather than $3/2$; the question is only what has been proved.

A The factorial-moment bound for the number of cycles

We derive, from the joining direction alone, the bound used in §4. Take $F = \emptyset$ (a uniform Latin square L of order n, so $m = n$; the argument uses only flips, switches and the identity $|J| = 2|X_A|$ of §4, so it applies in $\Omega(F)$ with m in place of n, but we state it for $F = \emptyset$). Let N_ℓ be the number of ℓ-cycles of ω and $\kappa = \sum_\ell N_\ell$. The two facts used are those behind $|J| = 2|X_A|$: for columns j_1, j_2 in different ω-cycles C_1, C_2, the flip at $\{j_1, j_2\}$ (the trade of the δ_{j_1,j_2}-cycle through row 2) is defined when $\{j_1, j_2\}$ is an A-pair and joins C_1, C_2 into one cycle of length $|C_1| + |C_2|$ with $\omega'^{|C_2|}(j_1) = j_2$ (Lemma 3.3), the other cycles being unchanged; and when $\{j_1, j_2\}$ is a B-pair, the switch along C_2 (we always switch the cycle of the second column, rather than the one selected by Definition 3.6; any fixed rule serves) inverts C_2 in place, leaves every other cycle unchanged, and makes $\{j_1, j_2\}$ an A-pair (Lemma 3.7). The backflip at $\{j_1, j_2\}$ in the joined square returns the square before the flip, so joining is at most two-to-one onto its image (the original square or its switch).

(i) $\mathbb{E}[\ell N_\ell] \leq 2n/(n - \ell)$ *for* $\ell < n$. For $\beta \neq \ell$, count instances $(L, \{j, j'\})$ with j on an ℓ-cycle and j' on a β-cycle: $\ell N_\ell \, \beta N_\beta$ per square; joining maps them at most two-to-one into the instances $(L', \{j, j'\})$ with $\omega'^{\beta}(j) = j'$ on an $(\ell + \beta)$-cycle, of which there are $(\ell + \beta) N_{\ell+\beta}$ per square (this is $|J| \leq 2|X|$ for the split (ℓ, β)). So $\mathbb{E}[\ell N_\ell \, \beta N_\beta] \leq 2\mathbb{E}[(\ell + \beta) N_{\ell+\beta}]$, and for $\beta = \ell$ the same count with the $\ell^2 \binom{N_\ell}{2}$ instances on unordered pairs of distinct ℓ-cycles and the $\ell N_{2\ell}$

unordered targets gives $\mathbb{E}[\ell^2 N_\ell(N_\ell-1)] \le 4\ell\mathbb{E}[N_{2\ell}]$. Summing over $\beta \ne \ell$, using $\sum_\beta \beta N_\beta = n$, and adding the $\beta = \ell$ term, $\mathbb{E}[\ell N_\ell(n-\ell N_\ell)] + \mathbb{E}[\ell^2 N_\ell(N_\ell-1)] \le 2\mathbb{E}\big[\sum_{m'\ge\ell+2} m' N_{m'}\big] \le 2n$, and the left side is $n\mathbb{E}[\ell N_\ell] - \ell\mathbb{E}[\ell N_\ell]$.

(ii) $\mathbb{E}[(\kappa)_k] \le 2^{k-1}\big(4+k^2/\log n\big)(\log n)^k$ *for every* $k \ge 1$. Count instances $(L, (C_1, j_1), \dots, (C_k, j_k))$ with $C_1, \dots, C_k$ pairwise distinct cycles of lengths $\ell_1, \dots, \ell_k$ and $j_i \in C_i$: there are $\ell_1 \cdots \ell_k$ per ordered k-tuple of distinct cycles with these lengths. Join C_1 and C_2 at $\{j_1, j_2\}$ (switching C_2 first if the pair is a B-pair), then join the result with C_3 at $\{j_1, j_3\}$, and so on, always switching the cycle of the new column when needed, so that the cycle containing j_1 is never inverted; after $k-1$ steps all j_i lie on one cycle of length $m' = \sum \ell_i$ at positions relative to j_1 determined by $(\ell_1, \dots, \ell_k)$, every cycle other than $C_1, \dots, C_k$ is unchanged, and the composite map is at most 2^{k-1}-to-one: the final square together with j_1 and $(\ell_1, \dots, \ell_k)$ determines the positions of $j_2, \dots, j_k$ on the joined cycle: a flip at $\{j_1, j\}$ changes ω only at j_1 and j, inserting the cycle of j after j_1, so after all joins j_i is at distance $\ell_i + \ell_{i+1} + \cdots + \ell_k$ from j_1 (switching a cycle before it is joined does not affect this); the backflips at $\{j_1, j_k\}, \dots, \{j_1, j_2\}$ in turn then return the squares before each join, and the only ambiguity is whether the cycle of j_i was switched before the $(i-1)$-st join, two choices for each of the $k-1$ joins. Hence the number of instances with lengths (ℓ_i) is at most $2^{k-1}\sum_{L'} m' N_{m'}(L')$, i.e. $\mathbb{E}[\ell_1 \cdots \ell_k\, T(\ell_1, \dots, \ell_k)] \le 2^{k-1}\mathbb{E}[m' N_{m'}]$ with T the number of ordered k-tuples of distinct cycles with these lengths. Summing over (ℓ_i), since $\sum_{(\ell_i)} T = (\kappa)_k$,

$$\mathbb{E}[(\kappa)_k] \le 2^{k-1} \sum_{\ell_1,\dots,\ell_k \ge 2} \frac{\mathbb{E}[m' N_{m'}]}{\ell_1 \cdots \ell_k}\Big|_{m' = \ell_1 + \cdots + \ell_k}.$$

Split at $m' \le n/2$, where $\mathbb{E}[m' N_{m'}] \le 4$ by (i) and $\sum_{m' \le n/2} \sum_{\sum \ell_i = m'} \prod \ell_i^{-1} \le (\sum_{\ell \le n/2} \ell^{-1})^k \le (\log n)^k$, and $m' > n/2$, where some $\ell_i \ge m'/k$, so $\sum_{\sum \ell_i = m'} \prod \ell_i^{-1} \le k \cdot (k/m')(\log n)^{k-1}$ and $\sum_{m' > n/2} \mathbb{E}[m' N_{m'}]\, k^2 (\log n)^{k-1}/m' \le k^2 (\log n)^{k-1}$, because at most one cycle is longer than $n/2$.

(iii) Tails. Markov's inequality on $(\kappa)_k$: with $k = \lceil \log n \rceil$ and $K = A \log n$, on $\{\kappa \ge K\}$ one has $(\kappa)_k \ge ((A-1)\log n)^k$, so $\Pr[\kappa \ge A \log n] \le (7 + \log n)(2/(A-1))^{\log n}$, which is $O(n^{-2}\log n)$ for $A \ge 16$; with $k = \lceil \sqrt{n} \rceil$ and $K = 9\sqrt{n}$, $(\kappa)_k \ge (8\sqrt{n})^k$ on $\{\kappa \ge K\}$ and $4 + k^2/\log n \le 2n/\log n$ for large n, so $\Pr[\kappa \ge 9\sqrt{n}] \le 2^{k-1}(2n/\log n)(\log n)^k/(8\sqrt{n})^k \le n\,(\log n/(4\sqrt{n}))^{\sqrt{n}}$ for large n, as used in §4.

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